ICCS206 Discrete Mathematic: Homework 9

Due on Tuesday, March 24, 2026

Pakawut Jiradilok Section 1

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tikz
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Problem 1: Minesweeper Theorem

Given any rectangular Minesweeper board, show that the sum of all hint numbers equals the sum of all hint numbers on the complement board (bombs and empty cells swapped).

Solution

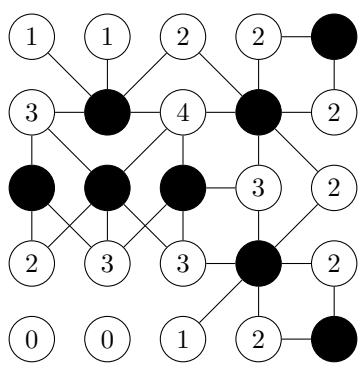
using the given minesweeper configuration as an example,

1	1	2	2	💣
3	💣	4	💣	2
💣	💣	💣	3	2
2	3	3	💣	2
0	0	1	2	💣

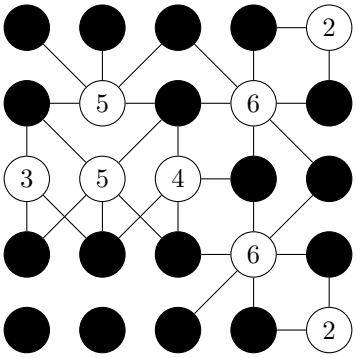
💣	💣	💣	💣	2
💣	5	💣	6	💣
3	5	4	💣	💣
💣	💣	💣	6	💣
💣	💣	💣	💣	2

we'll see that the summation of mines indication numbers are the same on the first configuration and it's reversed counter-part, equals to 33.

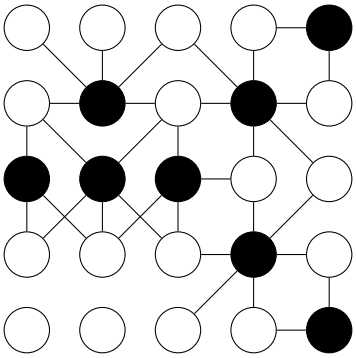
as the indication numbers are directly ties to the bomb (no relation to adjacent nodes of the same kind), we can write the first configuration as a graph as



and for the complement graph



and as it's a complement of the first configuration, obviously, the edges remain the same, and there's a clear separation between the initial bomb and indicator node, and the complemented ones. we could ignore the indicator numbers and write the final graph out as

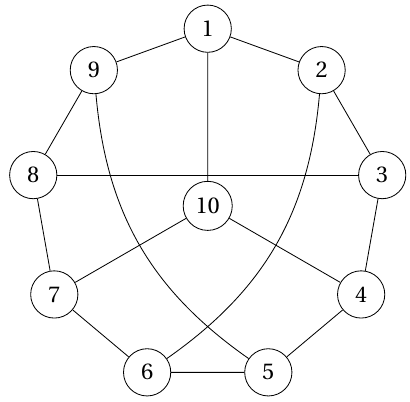


by the way we construct this graph, there's no way that an edge will connects an indicator node to another indicator node, or a bomb node to another bomb node.

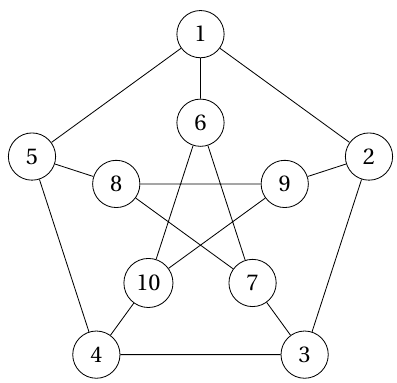
so every edge connects an indicator to a bomb, so each edge will contribute one degree to both the summation of indicator degrees and bomb degrees — this explains why the combined degrees of the first and second graph is the same, and is the same as the number of edges.

Problem 2: Graph Isomorphism

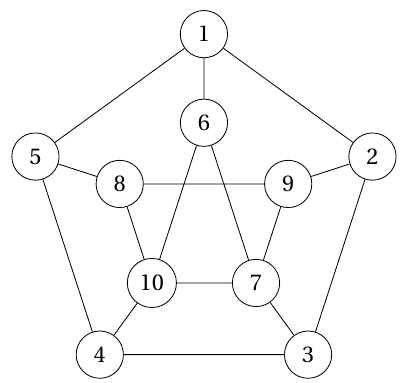
For the three graphs G_1, G_2, G_3 below determine which pairs, if any, are isomorphic. For each isomorphic pair provide an explicit isomorphism; otherwise briefly justify why they are not isomorphic.



G_1



G_2



G_3

Solution

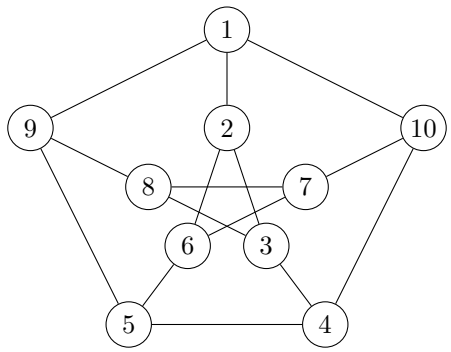
we first look at common properties of each graph. firstly, all the vertex in G_1 and G_2 have a degree of 3, but v_7 and v_{10} of G_3 got a degree of 4, so there's no way G_3 is isomorphic to G_1 and G_2, and we can just ignore it.

then, we check the combined degrees of G_1 and G_2, we got

$$\sum \text{degree}(v_1) = \sum \text{degree}(v_2) = 30 \quad \text{where } v_1 \in G_1, v_2 \in G_2$$

I supposed there's better way to check for isomorphism, but in this case, there's only 10 nodes, so I just look at some pattens and guess the placement; try substitude G_1 into G_2 while remain the properties of G_1.

after a few tries, i got



so

$$\varphi : V(G_1) \to V(G_2)$$
$$\varphi = \{\, 1 \mapsto 1,\ 2 \mapsto 10,\ 3 \mapsto 4,\ 4 \mapsto 5,\ 5 \mapsto 9,\ 6 \mapsto 2,\ 7 \mapsto 3,\ 8 \mapsto 8,\ 9 \mapsto 7,\ 10 \mapsto 6 \,\}$$

therefore

$$\boxed{G_1 \cong G_2}$$

Problem 3: Meeting Scheduler

We need to schedule executive meetings without conflicts (each person attends at most one meeting at a time). Each meeting lasts one hour.

Meeting	Members
A	{Smith, Jones, Brown, Green}
B	{Jones, Wagner, Chase}
C	{Harris, Oliver}
D	{Harris, Jones, Mason}
E	{Oliver, Cummings, Larson}

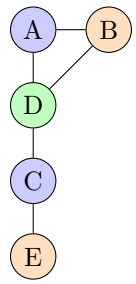
What is the minimum number of hours needed and how should the meetings be scheduled?

Solution

draw out venn-euler's diagram of the meetings, we'll see that

A overlaps with B, D
B overlaps with A, D
C overlaps with D, E
D overlaps with A, B, C
E overlap with C

we then construct the graph of meetings, with respect to the overlaps, and color it as



so, the schedule would be

hour	**meeting**
1	A, C
2	B, E
3	D

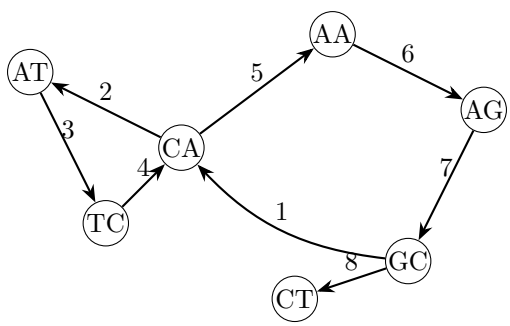
and it'll take at least $\boxed{\text{3 hours}}$ to complete.

Problem 4: DNA Sequence

Shotgun sequencing produced the subsequences $\{\text{CAT}, \text{TCA}, \text{AAG}, \text{GCT}, \text{GCA}, \text{ATC}, \text{CAA}, \text{AGC}\}$. What is the original sequence?

Solution

draw it out as a directed graph of



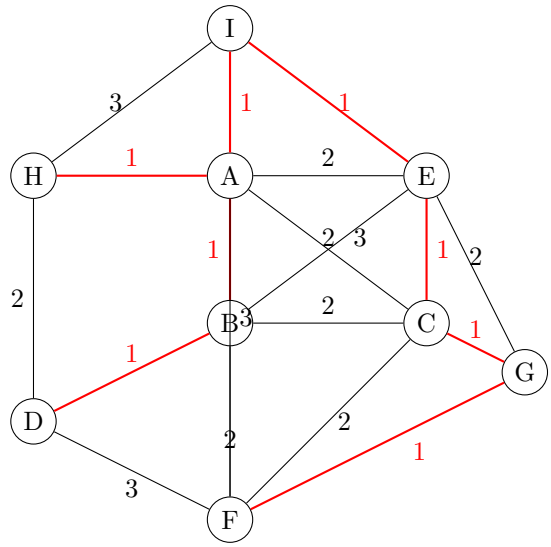
to find an euler walk on this directed graph, we start with GC, as it's one of the two vertices with odd degree (another one is CT, that won't work), so the original sequence is $\boxed{\text{GCATCAAGCT}}$.

Problem 5: Yoyo Distribution Network

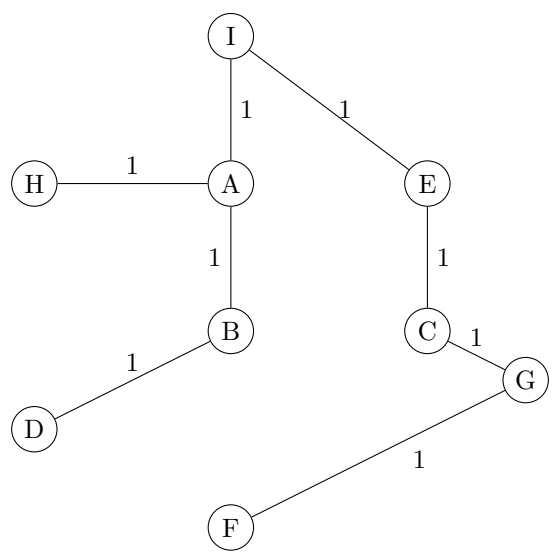
As CEO you must ship yoyos between distribution centers at minimum total cost. The network and edge costs are shown below (note AC costs 2, AD costs 1, and EB costs 3). Find the optimal routing plan.

Solution

we have to find a minimum spanning tree for the shortest distribution route, as shorter distance means less traveling time. to find a MST, we use Kruskal's algorithm: by repeatedly picking the smallest edge that doesn't cause a cycle in the graph.

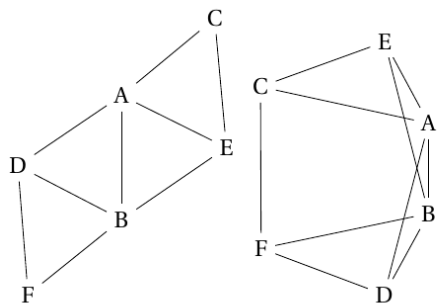


and, we'll be left with a MST with combined weight of $\boxed{8}$ — the MST could vary, but the weight will remain the same, and it could look something like this



Problem 6: One-Stroke Game

For each graph below determine whether it can be drawn in one stroke (an Euler trail). If so, exhibit the stroke order; otherwise explain why it is impossible.



Solution

Theorem. *Euler walk exists if and only if the graph is connected and there are exatly 0 or 2 odd degree vertex. Morover, if there are 2 odd degree vertices then, the Euler walk must start and end at the 2 odd degree vertices.*

say, the graph on the left is G_1; G_1 got 2 vertex with odd degree: D and E; so there exist an euler walk, and surely, the graph can be drawn with a single stroke as $\boxed{\text{DF, FB, BD, DA, AB, BE, EA, AC, CE}}$ path.

for the right graph, G_2, as there're 4 vertices with odd degrees: C, D, E, F; according to euler's theorem, the graph $\boxed{\text{does not has an euler walk}}$